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Determination of growth angles, wetting angles, interfacial tensions and capillary constant values of melts

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Abstract

This article explores the Voronkov approach to the problem of crystal surface formation in the crystal pulling process. The limits of fundamental condition of the growth angle (ε) constancy are discussed. Experiments and calculations are based on the shape of the small crystallized melt drops at the crystallization front after separation of the bulk crystal from the melt. The possibility of determination of the growth angle and the wetting angle (φ_{mo}) of the crystalline surfaces is shown. A simple method for determination of the melts capillary constant (a) and the melt surface tension σ_{LG}, based on measurements of the maximal meniscus height at the time moment of its breaking, is discussed.

Keywords: A1. Capillary constant; A1. Crystal surface formation; A1. Growth angle; A1. Surface energy; A1. Wetting angle; A2. Czochralski; A2. EFG

1. Introduction

This article presents an attempt at verification of Voronkov model [1–7] of crystal surface formation using separation drops at the crystallization front. His theory is applicable to the Czochralski, Stepanov, EFG, Verneuil, floating-zone methods and to whisker growth by the vapor–liquid–solid (VLS) mechanism. The importance of the growth angle value has become apparent from the modeling of the crystallization dynamics and the development of diameter control systems for growing crystal with weight control [8–11].

The problem of the lateral crystal surface formation consists in the determination of the orientation of a new external surface element of the crystal for a defined position of the melt surface adjacent to the three-phase line (TPL). For example, in the Cz method the melt surface orientation depends on the shape and the height of the TPL L (Fig. 1). Originally Voronkov used a pure phenomenological approach to this problem and his main approximation in Ref. [1] are: (1) rapid crystallization kinetics on all crystal front surface and (2) isotropic front and external crystalline surfaces in vicinity TPL. The assumption (1) allows one to neglect the small deviation from the equilibrium temperature during crystallization. The growth front is always curved in the

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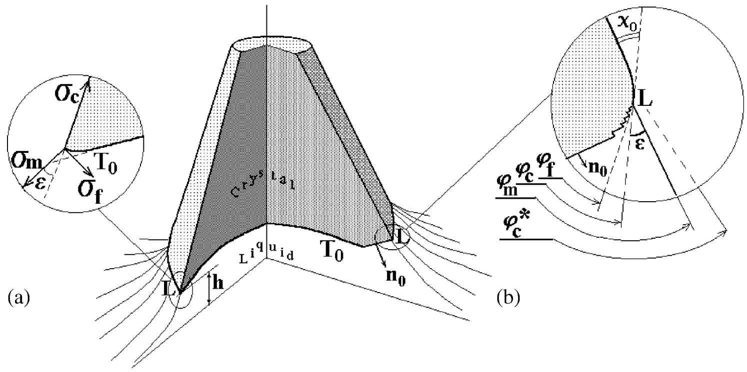


Fig. 1. The typical situation near the three-phase line. The section of the Cz crystal is shown as an example. Its growth front is concave and is intersected by a crystalline facet (on the right side). (a) The vicinity to the three-phase line L, far from the facet, (b) the vicinity of the facet with orientation $\mathbf{n}_0$. The growth front curvature at line L must be in equilibrium in accordance to Bolling–Tiller rule [12]. The crystalline steps accumulate adjacent to the TPL. Note, in case (b) the lateral crystal surface is curved. The angle χ_0 between the lateral surface position of the TPL and its final position far from L is determined by Eq. (2.2) ([3], see also Section 2). In case of (a) the angle $\chi_0 \to 0^\circ$.

vicinity of L and do not coincide with an isothermal surface T_0 (Fig. 1a) as the result of Gibbs–Thomson effect. This effect was described earlier in paper of Bolling and Tiller [12].

According to approximation (2) the vectors of the free surface energies σ_{SG} and σ_{SL} have only one component, lying in the interface surfaces, which coincides with the tangent to the phase boundary at a point on the TPL. Below we designate σ_{LG}, σ_{SG} and σ_{SL} as σ_m, σ_c and σ_f respectively. We also define $\mathbf{n}_i$ and $\mathbf{e}_i$ as the normal and tangent vectors to the surfaces $i=$ m, c, f. The idea of the analysis in Ref. [1] is: for any arbitrary shift of TPL during the crystallization (Fig. 2), the difference dF between change of the free energy of the crystal element, containing a unit of the TPL length and the produced work, should be nonnegative ($dF \geqslant 0$). If we define the interface orientations by the angles $\varphi_i, i=$ f, m, s, the free energy difference is $\delta F = \delta s[\sigma_c^2 - \sigma_m^2 \cos(\alpha) - \sigma_f^2 \cos(\alpha - \varphi_f)]$, where α is the arbitrary angle of displacement δs of line L (Fig. 2).

For the case of "incomplete wetting" of the lateral crystal surface by its own melt (when the length of $\boldsymbol{\sigma}_i$, $i=$ f, m, s vectors agree with the triangle inequalities) the direction of equilibrium displacement of the TPL (for which $dF = 0$) is equal to $\alpha = \varphi_C - \pi$. This angle is named as the growth angle ε, and its value is

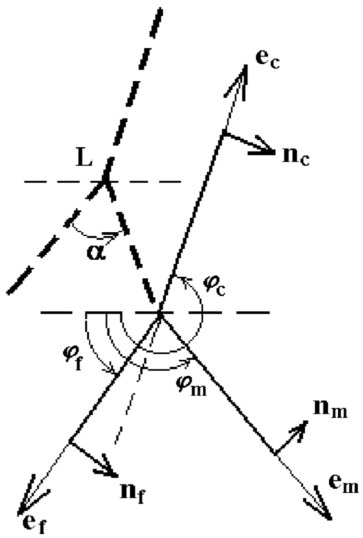


Fig. 2. Displacement of the TPL at an arbitrary angle α in the vertical plane of the crystal section (see Fig. 1). An equilibrium displacement of the TPL line L, at which the free energy difference $dF = 0$, is possible only at the angle $\alpha = \varphi_C - \pi$ [1]. Here, we neglect by the curvature of the phase-boundary surfaces for the picture scale.

determined by [1]

$$\varepsilon = \text{arc}\cos[(\sigma_m^2 + \sigma_c^2 - \sigma_f^2)(2\sigma_c\sigma_m)^{-1}]. \qquad (1.1)$$

Voronkov noted [1] that for the obtained angles ε, φ_f and φ_s the sum $\Sigma\sigma_i = 0$ at any point of the

TPL. In other words the triangle of surface forces, acting on the TPL, is closed during crystallization (see Fig. 19b).

In practice this case of "incomplete wetting" is the most common. However, also rarely, there is the case of "complete wetting" of the crystal by its own melt. For example, for the crystals of Cu, $LiNbO_3$, $LiTaO_3$ and ice, the experimentally determined growth angle $\varepsilon = 0°$ [13–16]. In this case, the triangle inequality is not fulfilled and the decreasing of a crystal surface becomes the thermodynamically preferable process ($dF < 0$). Voronkov [1] supposed that, on the lateral crystal surface, there is a microscopic fluid film, into which the melt flows. He showed that there should be some critical film thickness at which the change of phase takes place. In this case the growth angle will be zero ($\varepsilon = 0$). In later works [3–7] this concept was replaced by surface mass transfer due to diffusion.

The *lateral crystalline surface* can be curved in the vicinity of the TPL like the growth front due to the surface mass transfer. Such Gibbs–Thomson distortion of the lateral surface in the vicinity of TPL is especially notable if the singular facet adjoins line L (Fig. 1b). The singular facet must draw back from the isotherm T_0 into the more cold region for generation of the necessary rate of two-dimensional nucleation. The difference of chemical potential between solid and fluid phases produces of the atom stream from the melt along the lateral crystal surface. As the result the crystalline surface curved. The angle χ_0 between an initial position of the lateral crystal surface at the TPL and its final position (far from L) is equal (Fig. 1b) [3]

$$\chi_0 = KV^{-1/3}\, dT. \tag{1.2}$$

Here dT is the supercooling of the melt at the TPL, V is the velocity of L, and K is a coefficient depending on the speed of surface self-diffusion. The visual angle between the tangent to the melt meniscus and crystal surface far from line L (Fig. 1b) was termed by Voronkov as an "effective growth angle" $\varepsilon*$. It is equal to

$$\varepsilon* = \varphi_m - \varphi_c* = \varphi_m - \varphi_c - \chi_0 = \varepsilon - \chi_0. \tag{1.3}$$

In Refs. [3–7] it was shown that the effective growth angle ε^* can be negative. But the growth angle ε between the tangent to the meniscus and curved crystal lateral surface at the TPL (Fig. 1) varies very weakly due to smallness of magnitude of the angular derivative σ_i' of crystalline (i= c, f) surfaces free energy (see Section 3).

Periodic radial instability of Si whiskers during the VLS process [17] confirms this approach. In the whisker base there is always a cone (Fig. 3a), growing likewise a melt drop at the growth front (see Section 3). At the end of this cone formation we can always see the first oscillation in the whisker diameter. The following oscillations in crystal-melt system are connected with the dimensional dependence of the effective angle ε^* and appears when the Si crystal diameter is less than 1 μ. Effective growth angle ε^* (Fig. 3b) depends on the crystal radius and supercooling, which is determined by supersaturation in the solution-melt drop at the whisker apex. The oscillations in the diameter of submicron Si whiskers are explained by the appearance of limit cycles in the crystallization system, which was described in Ref. [18] as the dynamic system of the second order (Fig. 3).

The theory, developed in works [1–7], provide a detailed description of the external morphology of real crystals being pulled from the melt. The orientation of the lateral crystal surface (Fig. 1b) is determined by the equation

$$\varphi_c^* = \varphi_m - \varepsilon + \chi_0 \tag{1.4}$$

and depends not only on values of free energies of boundary surfaces and orientation of the seed, but also on many other factors (the temperature

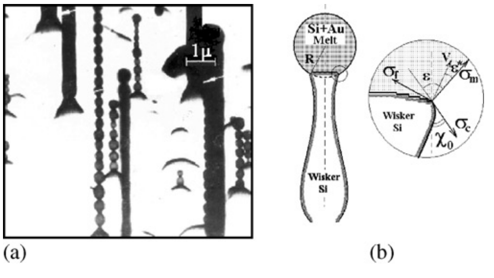


Fig. 3. (a) Photo of Si submicron whiskers [17]. (b) The diameter oscillations of Si whiskers are explained by positive feed back and appearance of limit cycles on the phase-plane portrait of the second order crystallization dynamic system [18].

gradients, coefficients of surface mass transfer, kinetic coefficients at the growth front, capillarity, etc.). Generally, the angle between the tangent to the meniscus and the new lateral surface of the crystal (far from the TPL) is a variable value and is determined by Eq. (1.3). The presence of singular facets in the vicinity to the TPL produces variations of this angle in different methods of crystal growth [2,6,7]. The angle ε is determined by the free energies of boundary surfaces. It varies very weakly in case of nonsingular boundary surfaces near the TPL and can be considered as a fundamental constant of the crystal [2–7]. The angle ε is especially important in the mathematical modeling of the Cz method and design of feedback control of crystal diameter [9–11]. The conditions near the TPL can be considered in the isotropic approach along almost its full length in bulk crystal growth. This is because the singular facets are localized to a small section of the crystal perimeter by the high values of the temperature gradients.

2. The melt drop on a growth front and the modern approach to the problem of lateral crystal surface formation

The formalism of the modern Voronkov theory of crystal surface formation takes into account the crystalline surfaces anisotropy, surface mass transfer and is based on the use of three classical results:

- the Wulff theorem about the equilibrium crystal shape [4,5,19],
- the Herring theorems about equilibrium of several crystalline anisotropic phases at a junction line and free surface energies of such surfaces [20,21],
- the geometrical construction of Hoffman–Cahn, proposed in Ref. [22].

The equilibrium crystal form (at constant volume) is determined by the minimum of the first variation of the crystal free surface energy functional $\delta \int\int \sigma(n)\,\mathrm{d}s = \min$. The solution of this task is to find the envelop family of planes $P(\mathbf{n})$ which is described by the equation $(\mathbf{r} \cdot \mathbf{n}) = \sigma(\mathbf{n})$, where $\mathbf{r}$ is a

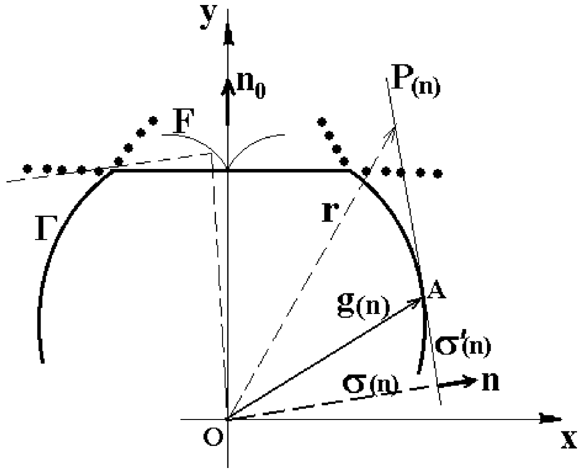


Fig. 4. Envelop Γ of planes $P_i(\mathbf{n}_i)$ is the section of the spatial Wulff figure. The stable crystalline surfaces P_i correspond to the convex (outside) portion of the envelope. Doted lines show the metastable orientations (surfaces) on the Wulff figure Γ. Planar parts on Γ correspond to the singular orientations (minimum on the polar diagram F).

vector from center O to the point on the plane $P(\mathbf{n})$ [4,5,19]. The geometric solution of the above equation is based on the using of the polar diagram and Wulff approach. The equilibrium crystal surface coincides with the envelop Γ of planes $P_i(\mathbf{n}_i)$ in Fig. 4 (Wulff figure).

The two *crystalline* interfaces may be singular, vicinal or weakly anisotropic. This surfaces are classified as stable, metastable and labile [4,5,19,23]. For the stable surfaces the second variation of the free surface energy functional is positive $\delta^2 \int\int \sigma(\mathbf{n})\,\mathrm{d}s > 0$. The stable orientation $\mathbf{n}$ must have the positive radius of curvature $\sigma(\mathbf{n}) + \sigma''(\mathbf{n}) > 0$ at the corresponding point of the Wulff figure. Here $\sigma''(\mathbf{n})$ is the second angular differential of the free energy $\sigma(\mathbf{n})$ [19]. The singular orientations correspond to planar parts on the envelope Γ (Fig. 4), since the angular differential of the polar diagram in this case $\sigma''(\mathbf{n}) = 0$. The length of such planar parts is determined by the magnitude of the jump $\sigma'(\mathbf{n}_0)$ [4,5,23]. The metastable orientations are indicated by dotted lines in Fig. 4. The labile orientations, with concave (outside) portions of Γ, are not shown in Fig. 4.

During crystal pulling from the melt only three surfaces intersect at the TPL. The general condition for equilibrium $i \geqslant 3$ of intersecting surfaces was defined by Herring. The variation of a free surface energy for small deflections of anisotropic surfaces is equal to $\delta F = -\sum(\sigma_i \mathbf{e} + \sigma_i^{/}\mathbf{n})\delta s$ [20]. So the well-known equilibrium ($\delta F = 0$) conditions at the intersection line may be written as the vector balance

$$\sum_i (\sigma_i \mathbf{e} + \sigma_i' \mathbf{n}) = 0. \qquad (2.1)$$

Originally this equation was used in articles [2,8] for the definition of the equilibrium position of phase boundaries during stationary crystallization, or in other words for the determination of the growth angle ε during stationary crystallization. The solution of two independent equations for the surfaces force and moments of this forces balance give expression (1.1) plus any correction term due to taking into account the angular derivatives σ_c' and σ_f'. Quantitative estimations allow one to neglect these terms due to their small values, but only if the crystalline surfaces at the TPL are not singular facets [8]. Problem of crystal surface formation in the common case, i.e. the presence of singular facets at the TPL, was investigated in Refs. [2–7] by the use of a graphical solution of the Herring equations (2.1) and the vector thermodynamics approach of Hoffman–Cahn [22].

The graphical solution is based on the vectors $\mathbf{g}_i(\mathbf{n}) = \sigma_i \mathbf{n} + \sigma_i' \boldsymbol{\tau}$, where i= c, f, $\boldsymbol{\tau}$ is the unit vector, situated on the plane P_i touching the Wulff figure at a point A (Fig. 4), $\mathbf{n}$ is unit vector, normal to the plane P_i, and $\sigma_i^{/}(\boldsymbol{\tau})$ is the angular derivative of free surface energy. Since a rotation of 90° of all the vectors in the Herring equations (2.1) does not destroy this balance condition at the TPL, it can be written as [4,5,22]

$$g_c(\mathbf{n}_c) = g_f(\mathbf{n}_f) + g_m(\mathbf{n}_m). \qquad (2.2)$$

An example of graphical construction (Cahn–Hoffman (a) and Voronkov (b) solution of Herring equations) for any equilibrium position of interface surfaces at the TPL is shown in Fig. 5.

Since always $\sigma_c > \sigma_f$, the Wulff figure Γ_f for the crystal front surface will be enclosed within the figure Γ_c. At the same time the figure Γ_m is the circle with its center on Γ_c or on Γ_f. The circle Γ_m center position is defined by the choice of azimuthal orientation of one of the crystalline surfaces. If the orientation of the crystallization front is fixed, the center of Γ_m is positioned on Γ_f. In this case, according to the Voronkov rule, the unique solution of Eq. (2.2) is determined by an intersection point of the figures Γ_m and Γ_c, which must be "entering" point into Γ_c if the movement around the circle Γ_m is counterclockwise [4,5]. Thus, as a result of "moving" of the Γ_m center around Γ_f, it is possible to construct the angular diagrams $\varphi_c(\varphi_m)$ and $\varphi_f(\varphi_m)$, which allows one to define the orientations of two surfaces by the use of the orientation of the third surface [4–7].

Experiments on the crystallization of melt drops at the crystallization front allow one to test some

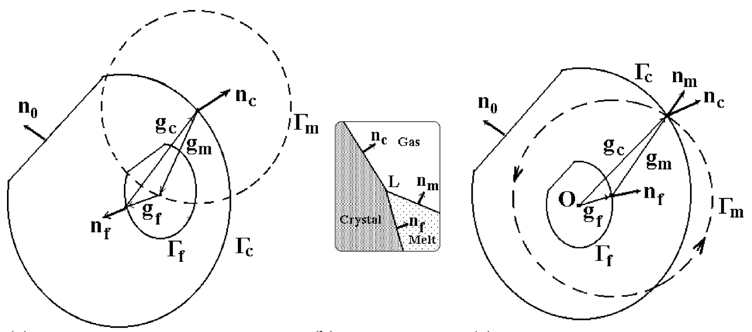


Fig. 5. Graphical solution of the Herring equation (2.1). (a) According to Cahn–Hoffman [22] and (c) according to Voronkov [4,5]. (b) Is the equilibrium arrangement of a interface surfaces in the vicinity of three-phase line L. In Voronkov graphical construction (c) the Wulff figures Γ_f and Γ_c have common center O. The vector $\mathbf{g}_m$ is equal to the radius of the circle Γ_m, which is the section of the spherical Wulff figure for the isotropic melt surface. The Cahn–Hoffman approach differs from Voronkov approach also in choice of the direction of the normals to the interface surfaces growth front and melt (see Ref. [5]).

conclusions of the Voronkov theory. After separation of the growing crystal from the melt, shrinkage of the remaining liquid film into the small melt drop takes place because melt wetting of the crystalline surface is incomplete. The formation of the drop is completed after reaching equilibrium of the interfacial forces in the plane of the front crystallization (see Fig. 19b). The vertical component of the sum of interfacial tensions is compensated by the solid phase deformation. The stages of crystallization of a Ge drop on an isotropic, slightly convex, front are shown in Fig. 6. Finally, a symmetrical cone-shaped vertical knoll (denoted further for simplicity as the cone) occurs at the front, having a complex shape of lateral surface. The pioneer in using such cones for

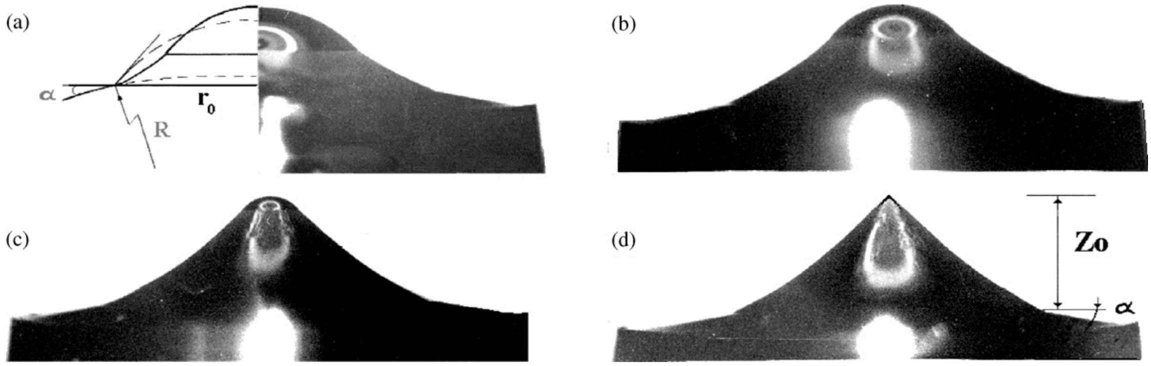


Fig. 6. Photo of a Ge drop crystallization on a isotropic growth front after separation of the growing crystal from the melt. The image is inverted with respect to direction of the crystal pulling. The macroscopic growth front curvature can be determined from the angle α.

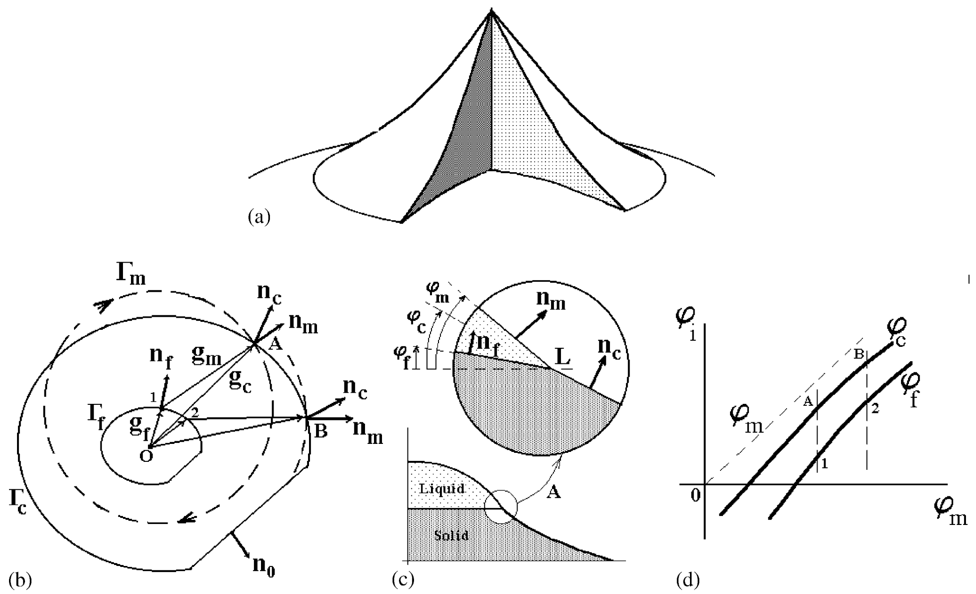


Fig. 7. (a) Schematic of the cone on an isotropic growth front (see Fig. 6). (b) Graphical solution of the equilibrium equations (2.2), based on Voronkov construction. (c) Position of the surfaces in the vicinity of the three-phase line L during drop crystallization, which corresponds to point A in the figure Γ_c. (d) The angular Voronkov diagrams. The changes of slope angles to the liquid phase at the TPL during the melt drop crystallization corresponds to "shifting" from point 1 to 2 in the figure Γ_f. Accordingly, the changes of slope angles to the solid phase vary from point A up to point B in figure Γ_c.

determination of the growth angle was Voronkov [24]. But now we consider the cones formation in light of already briefly discussed positions.

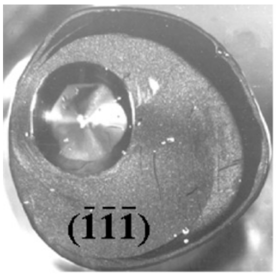
We emphasize three different cases of the initial location of the drop (and cone) at the crystal front:

(I) The base of the melt drop (and cone) is completely located on the isotropic surface of the growth front (Figs. 6 and 7a).
(II) The base of the melt drop is completely located on the singular facet at the growth front (Fig. 8).
(III) The base of the drop places both on an isotropic part of front, and on the planar singular facet (Fig. 10).

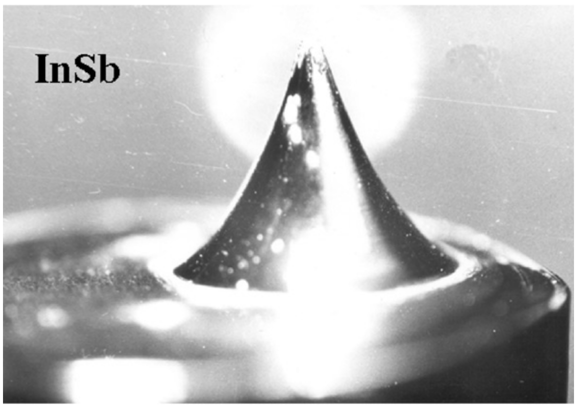
In *case* I, a wonderfully symmetric cone appears after crystallization of the melt drop on an isotropic growth front (see Figs. 6 and 7a). It is important to note, that the drop crystallization takes place without the involvement of the singular surfaces at the growth front and any singularities on the cone lateral surface. This fact allows us to carry out the geometrical construction shown in Fig. 7b.

The Wulff figures Γ_i, where $i = c, f, m$, we can represent only qualitatively since the values of interphase energies and the shape of the *polar diagrams presently are unknown*. However, it is possible to take into account some general postulates including the fact that, in this case, singular orientations $\mathbf{n}_0$ do not take part in cone shaping. In Fig. 7b the geometrical construction of the equilibrium positions of boundary surfaces by use of the Voronkov approach for case I is shown. The center of the circle Γ_m "moved" clockwise from point 1–2 along the Wulff figure during crystallization of the drop on the cone apex. The closed triangle (AO1) composed of vectors $\mathbf{g}_i$ represent the graphical solution of Eq. (2.2). The top of this triangle "turns" clockwise also along Γ_c from position A to B. In Fig. 7c are shown the equilibrium positions of interfaces in the vicinity of the three-phase line L, which corresponds to point A in Fig. 7b (intermediate stage of the drop crystallization on the cone apex Fig. 7c). The crystallization front near L should be curved according to the Gibbs–Thomson effect. The tangent slope φ_f to the growth front in the vicinity of L in Fig. 7c is determined by the normal $\mathbf{n}_f$ in point 1 to the figure Γ_f (i.e. equal to the tangent slope to Γ_f at this point 1). In Fig. 7d are shown the angular diagrams $\varphi_c(\varphi_m)$ and $\varphi_f(\varphi_m)$ for case I. The growth angle is $\varepsilon = \varphi_m - \varphi_c$ (Fig. 7c and d). According to Fig. 7d the growth angle ε is almost constant. This conclusion follows because the sectors 1–2 and AB on the Wulff figures Γ_f and Γ_c can by approximated by circles. Γ_f should be similar to Γ_c and moreover the distance between Γ_f and Γ_c is practically constant. It is necessary to interpret the Voronkov conclusion about the constancy of the growth angle in the case of isotropic interfaces [1–5] in this sense.

In case II the melt drop base is initially situated completely on the singular surface at the crystal front. The InSb cone, formed by crystallization of



(a)



(b)

Fig. 8. (a) View from above the InSb cone, the base of which is completely located on the planar facet $(\bar{1}\,\bar{1}\,\bar{1})$. This facet occupies almost all the surface of the crystal front; (b) side view of this cone. A lateral surface of the cone is smooth, without any facet. All horizontal sections of the cone are circles.

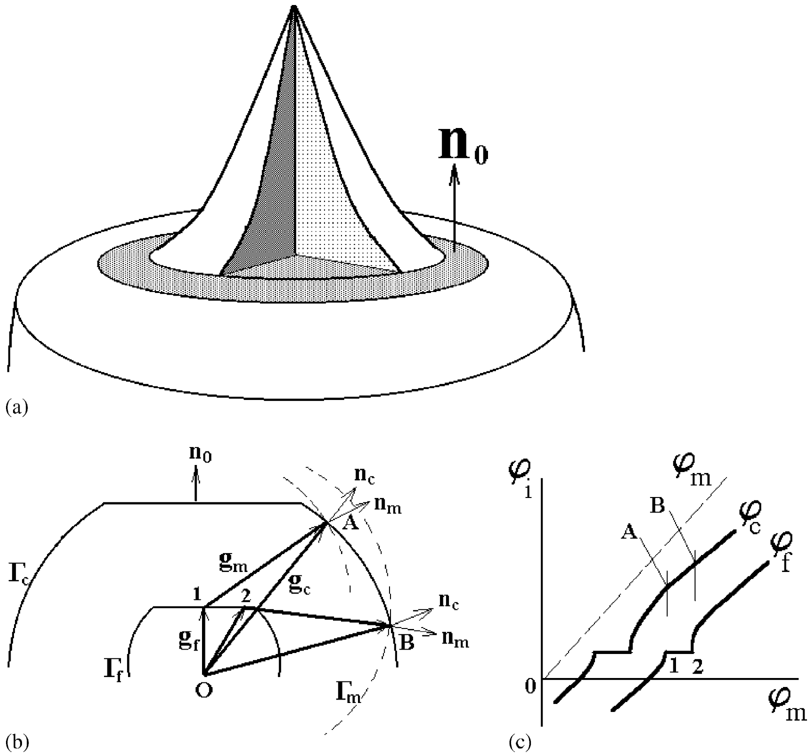


Fig. 9. (a) Schematic of an InSb cone (Fig. 8) on the crystal front, faceting by the flat surface **($\bar{1}\,\bar{1}\,\bar{1}$),** (b) the graphical solution of Eq. (2.2), (c) the angular Voronkov diagrams. During melt drop crystallization the growth front remains planar from the beginning (point 1 on the curve $\varphi_f(\varphi_m)$) to the end of the cone formation (point 2 on a curve $\varphi_f(\varphi_m)$). The slope of the lateral cone surface varies from point A up to point B on the curve $\varphi_c(\varphi_m)$. The region AB is situated far from the region of "external" faceting, i.e. the horizontal on the curve $\varphi_c(\varphi_m)$. Therefore, the lateral cone surface is smooth and has no features (see Fig. 8).

the melt drop on the planar facet **($\bar{1}\,\bar{1}\,\bar{1}$)**, is shown in Fig. 8.

This cone is shown schematically in Fig. 9a. The geometrical solution of the Herring equations for this case II is shown in Fig. 9b.

Note, the lateral cone surface is smooth and has no facetting tracks in this case. That means the orientation of lateral surface from the cone base (point A in Fig. 9b) up to its apex (point B) does not coincide with any singular orientations $\mathbf{n}_0$. At the same time the solidified front is initially planar (having the orientation $\mathbf{n}_0$, point 1 on Γ_m), and it will remain planar up to the cone apex (point 2 on Γ_m). The vicinal surface of the growth front covered by the steps with small surface density and remains macroscopically planar during the drop crystallization. The constancy of the front orientation $\mathbf{n}_0$ keeps the curve $\varphi_f(\varphi_m)$ horizontal (region 1–2 in Fig. 9b) when the slope angle to the melt drop φ_m incremented (Fig. 9c). In case II the flat horizontal region of curves $\varphi_c(\varphi_m)$ and $\varphi_f(\varphi_m)$ are situated separately one after another. It is important, that the region A–B at the curve $\varphi_c(\varphi_m)$ is smooth (see Fig. 9c). But in contrast to the previous case I, the growth angle $\varepsilon = \varphi_m - \varphi_c$ changes a little from the beginning to the end of the melt drop crystallization. Why? Because the equilibrium triangle, composed of vectors $\mathbf{g}_i$, where $i = \mathrm{c, m, f}$, is "deforming" when it turn clockwise along the curve Γ_c from position A to position B (i.e. the length of vector **O1** in Fig. 9c is less than **O2** while the length of $\mathbf{g}_m$ and $\mathbf{g}_c$ are invariant). The calculated values of ε, obtained in experiments with InSb drops, confirm this conclusion (see Section 3, Table 1 and Fig. 14c).

Table 1
The angles φ_i, ψ_i and ε_i, obtained by use of the parabolic cone profile approximation (Fig. 13b)

Material	Z_i (mm)	r_i (mm)	φ_i (deg)	ψ_i (deg)	ε_i (deg)
Si (isotropic), $r_0 = 5.10$ mm, $Z_0 = 3.60$ mm, $\alpha = 0°$ (see Fig. 15e)	0.060	4.948	33.142	21.786	11.356
	0.360	4.246	35.812	24.509	11.303
	0.660	3.624	38.241	26.956	11.285
	0.960	3.061	40.506	29.186	11.320
	1.260	2.546	42.680	31.239	11.441
	1.560	2.069	44.873	33.143	11.730
	1.860	1.625	47.335	34.920	12.415
	2.120	1.208	50.848	36.584	14.264
	2.460	0.816	58.832	38.150	20.682
	2.760	0.444	90.123	39.627	50.496
InSb $(\bar{1}\,\bar{1}\,\bar{1})$, $r_0 = 2.03$ mm, $Z_0 = 2.80$ mm, $\alpha = 0°$ (see Figs. 14c and 15d)	0.060	1.848	65.231	40.295	24.936
	0.360	1.529	72.184	46.256	25.928
	0.660	1.267	78.350	51.487	26.863
	0.960	1.048	83.838	56.103	27.735
	1.260	0.862	88.678	60.194	28.484
	1.560	0.702	92.799	63.833	28.966
	1.860	0.566	95.943	67.085	28.858
	2.120	0.448	97.314	70.001	27.313
	2.460	0.346	94.038	72.628	21.410
	2.760	0.259	65.156	75.004	−9.848

The growth angle is defined as the difference $\varepsilon_{i=}\varphi_i - \psi_i$ between angles of the tangent slope to the meniscus $\varphi(r,z)$ and to the lateral cone surface $\psi(r,z)$ at the TPL for any time of the drop crystallization.

In case III the melt drop base is initially located simultaneously on singular and partly on isotropic crystal front (Fig. 10).

In this case III the lateral surface of the cone has some step, associated with melt slide along a planar singular front. However it is impossible to explain this step by the slide of melt along the front! It is necessary to take into account that the slope angle to the drop at the TPL is constantly increasing during the process of melt drop crystallization. Simple mechanical sliding of the melt along the crystal surface is possible if the slope to the melt at the TPL becomes less than the wetting angle during the drop crystallization. The temporal evolution of the cone (shown in Fig. 10) formation process is shown schematically in Fig. 11.

Note, that of the slope to the drop at the TPL is close to the wetting angle φ_{m0} at the initial moment of crystallization at the front surface with orientation $\mathbf{n}_0$. The angle of the tangent to the meniscus at the TPL will only increase during drop crystallization from point A to point C in Fig. 11 (also see Section 3). Therefore, simple mechanical sliding of the melt along the facet becomes impossible. Nevertheless, at some angle, greater than the wetting angle φ_{m0}, the direction of crystallization changes and lateral cone surface faceting at the singular orientation $\mathbf{n}_0$ occurs (at point B in Fig. 11). At last, the direction of crystallization changes once again in point C and the step appears on the lateral cone surface.

The beginning of cone formation corresponds to points 1 on Γ_f and A on Γ_c (Fig. 12a and b). Points 2 and D correspond to the termination of drop crystallization. In other words triangle O1A, composed from vectors $\mathbf{g}_i$ (i.e. the graphical solution of Herring's equations), "rotates" clockwise in this figure. Thus, the angle between vectors $\mathbf{n}_m$ and $\mathbf{n}_0$ increased continuously from point 1 to point 2. Before the beginning of drop crystallization, this angle is at its minimum and can be assumed to be the wetting angle φ_{m0} of the melt by the crystalline surface. The separation drop formation at the front ends at this slope angle to the meniscus φ_{m0} (Figs. 11 and 19b). After the

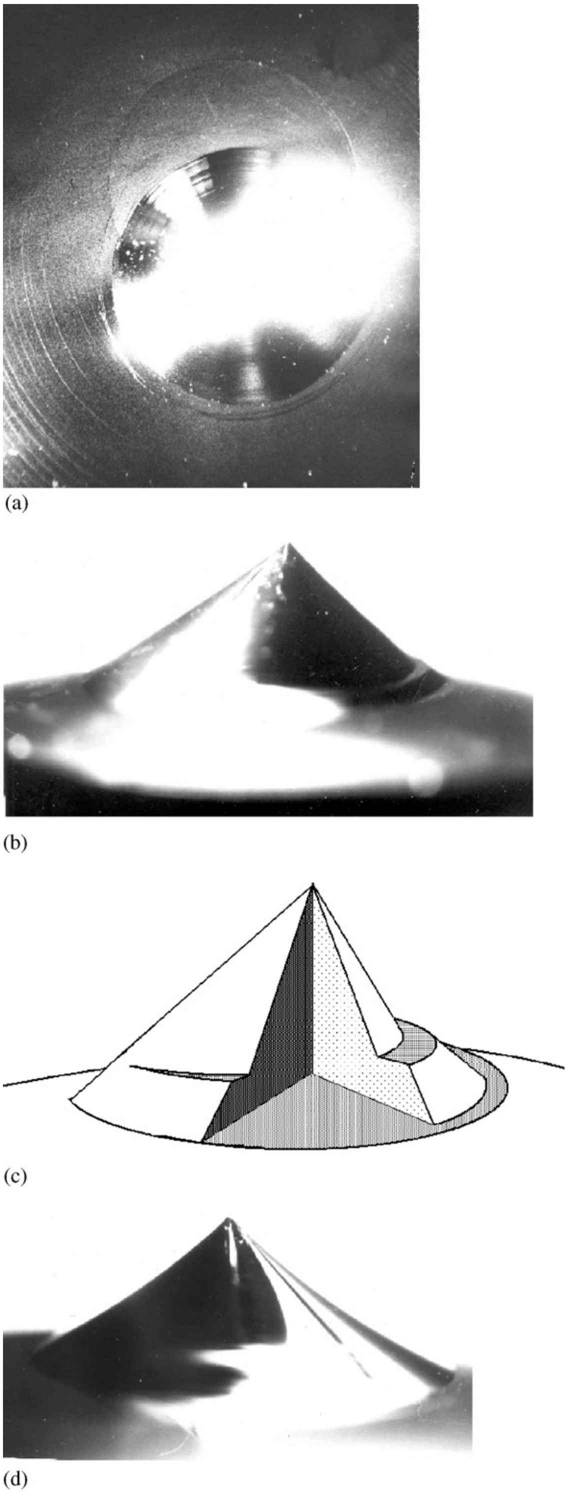


Fig. 10. (a) View from above of Si cone, the basis of which is located *simultaneously* on mirror-like facet surface and partly on isotropic crystal front, (b) side view of the cone in the direction of singular front, (c) the scheme of the cone with the steps on its lateral surface, (d) the side view on the cone in the direction of the isotropic front.

beginning of drop crystallization, a unique solution of Herring's equations is possible only at point A on the branch of *metastable* orientations of crystalline surfaces on Γ_c (Fig. 12a). These branches of metastable orientations (see Refs. [4,5]) are shown in this figure by the dotted line. Thus the orientation $\mathbf{n}_c$ of lateral cone surface near its base differ from the singular orientation $\mathbf{n}_0$. The angle between the planar front and tangent to the melt meniscus at the cone apex will increase during crystallization (see Section 3 of this article). Since the crystallization front remains macroscopically planar so, the circle center, having a radius $|\mathbf{g}_m|$, is shifted to the right from point 1 to point 2 on Γ_m. Therefore, at point B a new solution of Herring's equations occurs on the *stable* (but singular) part of figure Γ_c. Physically this means that the lateral cone surface changes orientation and, on the lateral cone surface, a planar facet with orientation $\mathbf{n}_0$ (which is favored energywise) appears. The macroscopic orientation $\mathbf{n}_0$ of the crystallization front and lateral cone surface "along the segment BC" do not change. Once again the transition from the metastable branch to the stable branch of the Wulff figure Γ_c (from point C in the direction of point D, Fig. 12) is favored energywise during *increase* of the tangent slope to the melt at the TPL (Fig. 12).

3. Verification of the growth angle constancy

For a small initial melt mass m_0 the influence of gravitation on its equilibrium shape can be neglected. The meniscus of the melt drop has the shape of a sphere segment and the tangent to it on a TPL before crystallization begins is equal to the equilibrium angle of wetting φ_{m0}. The crystallization of a drop for opaque semiconducting crystals begins not from the apex, but from its base at the crystallization front. In Fig. 6 enlarged freeze frames of the crystallization process for a **Ge** drop on an isotropic, slightly convex, growth front is shown. Finally, a cone-shaped, symmetrical vertical knoll with complex shape, called further

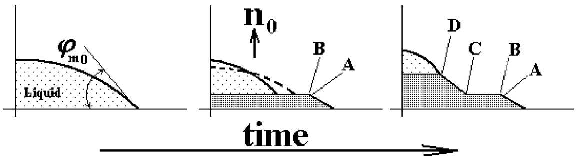


Fig. 11. Time scan of cone formation, which has a step on its lateral surface. The base of the melt drop (its right part) is placed initially on the facet with orientation $\mathbf{n}_0$. The initial slope to the drop is close to the wetting angle φ_{m0}. The dotted line shows the melt drop between points B and C. The tangent slope to the drop at the TPL increased steadily during crystallization. It is impossible to explain the formation of the step on the cone surface by the down of the melt along the crystallization front (see text).

for simplicity as the cone, occurs. The thermal environment of the crystal does not change, due to the small crystal displacement from the melt and small size of the crystallized drop. So it is possible to assume constancy of macroscopic growth front curvature during drop solidification. This requirement is fulfilled, at least on an initial site of crystallization of a cone not far from its base. The used approximations are always correct for drop crystallization on a planar singular surface, for example, in the case of InSb (see Fig. 8). Taking into account the initial macroscopic growth front curvature (see Fig. 6) does not cause any difficulties and is omitted here in the calculations. The scheme, represented in Fig. 13, is used for the following calculations.

The growth angle is defined as the difference $\varepsilon = \varphi - \psi$ between the angles of the tangent slope to the meniscus $\varphi(r)$ and to its lateral surface $\psi(r)$

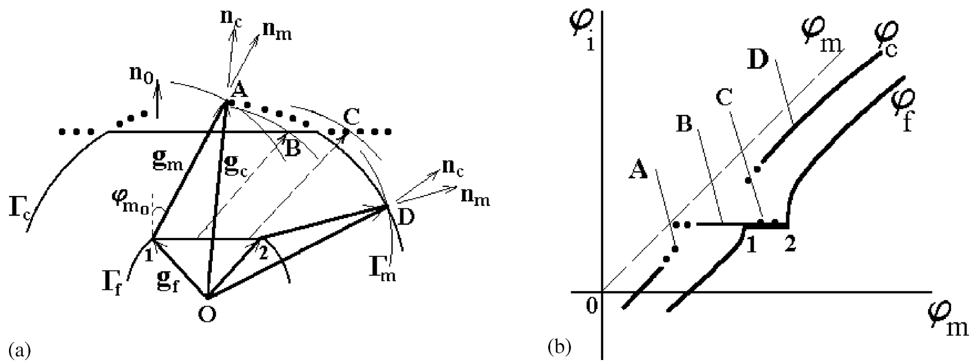


Fig. 12. (a) Geometrical solution of Eq. (2.4), (b) Voronkov angle diagram for the case of the appearance of a step on the lateral cone surface (see Figs. 10 and 11). The dotted line shows the branches on the Wulff figure Γ_c, corresponding to metastable surfaces of the solid phase [4,5,19]. The horizontal regions on the angular diagram $\varphi_f(\varphi_m)$ for the crystallization front and $\varphi c(\varphi m)$ for the cone surface correspond to a singular orientation $\mathbf{n}_0$. In this case III these regions partly coincide. The segment between points B and C on the angular diagram correspond to the planar surface of the step BC in Figs. 10 and 11 (see text).

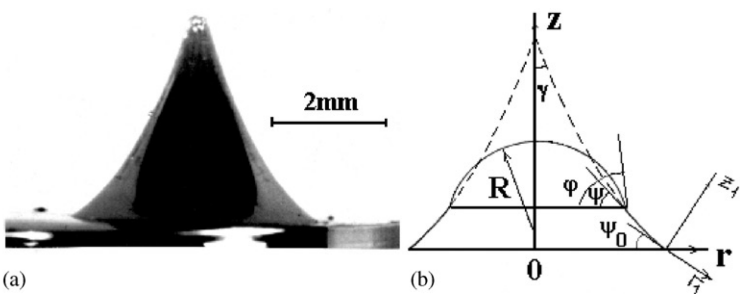


Fig. 13. (a) Photo of an InSb cone on a planar front $(\bar{1}\,\bar{1}\,\bar{1})$, (b) schematic, adopted in calculations.

at the TPL for any given instant of the drop crystallization. Note that no preliminary suppositions about growth angle constancy are made. The value and constancy of the growth angle ε should be determined from calculations of the cone shape. The testing of the growth angle constancy is reduced to the evaluation of crystallized drop parameters at any moment of the cone growth. The symmetry of the cone allows one to calculate the required angles. Only the densities of the solid (ρ_S) and the liquid (ρ) phases of the materials need be known. It is necessary to determine the equation of the cone profile.

It is obvious from Fig. 13 that the profile of the axisymmetric cone is well approximated by the parabola $z_1 = kr_1^2$ in the coordinate system (z_1, r_1), shifted to the edge of the cone and rotated by the angle ψ_0. Indeed, in this coordinate system, the equation $\mathrm{d}z_1/\mathrm{d}r_1 = -tg(\psi - \psi_0)$ is valid. For small angles, $\Delta\psi = \psi - \psi_0$, it is possible to restrict the Taylor expansion of $tg(\Delta\psi) = \Delta\psi + (\Delta\psi)^3/3 + \ldots$ to the first term $\Delta\psi = (\mathrm{d}\psi/\mathrm{d}r)r$. The integration of the obtained equation $\mathrm{d}z_1/\mathrm{d}r_1 = -(\mathrm{d}\psi/\mathrm{d}r_1)|_{r1=0}r_1$ give the equation $z_1 = kr_1^2$, where $k = -(\mathrm{d}\psi/\mathrm{d}r_1)|_{r1=0} = \mathrm{const}$. The accuracy of such an approximation is higher near the cone base and becomes worse at its apex. Parameter k and angle of the tangent to the cone ψ_0 can be determined by the use of cone photos. Note, we can take into account the known focal properties of a parabola in the measurements. In the coordinate system (r, z) with cylindrical cone symmetry, the transformation $z = z_1\cos(\psi_0) - r_1\sin(\psi_0)$ and $r = z_1\sin(\psi_0) + r_1\cos(\psi_0) + r_0$ finally gives the equation of the cone profile:

$$r(z) = kC^2\sin(\psi_0) + C\cos(\psi_0) + r_0, \tag{3.1}$$

where $C = tg(\psi_0)/2k - B$ and $B = [(tg(\psi_0)/2k)^2 + z/(k\cos(\psi_0)]^{1/2}$. Differentiating this equation, we find the angle of the profile curve slope for the cone at any point:

$$\psi(r) = -\mathrm{arctg}(Cctg(\psi_0) - 1/A), \tag{3.2}$$

where $A = (\cos^2(\psi_0) - 4k(r - r_0)\sin(\psi_0))^{1/2}$ and r_0 is the radius of the cone base. The second angle $\varphi(r)$, which is necessary for definition of the growth angle, is defined by the variable volume of the melt drop on the cone apex at intermediate moments of crystallization (Fig. 6), corresponding to cone heights $0 < z < z_0$, where z_0 is its complete height:

$$V = \rho_s\rho^{-1}\left(V_0 - \pi\int_0^z r^2(z)\,\mathrm{d}z\right). \tag{3.3}$$

Here $V_0 = m_0/\rho_s$ and $m_0 = \rho_s\pi\int_0^{z0} r^2(z)\,dz$. The melt meniscus has the shape of a sphere segment at all times of crystallization. Its current volume is $V = \pi h(h^2 + 3r^2)/6$. This expression can be presented as a cubic equation having as independent variable a function of the required angle φ,

$$f^3(\varphi) + 3f(\varphi) - 6V/\pi r^3 = 0, \tag{3.4}$$

where $f(\varphi) = (1 - \cos(\varphi))/\sin(\varphi)$. The discriminant of this equation is positive and allows one to calculate the roots of Eq. (3.4) by the well-known Cardan formulae (see Section 4). The results of numerical calculations of the angles ε, $\varphi(r, z)$ and $\psi(r, z)$ for the materials InSb, Ge and Si are shown in Table 1 and Fig. 14. The radius of the cone base r_0, its height z_0 and the crystal front orientation are shown in the first column of Table 1 for the case of Si and InSb crystals. The crystal fronts are planar in both cases ($\alpha = 0°$). The accuracy of calculations of the angles φ_i, ψ_i and ε_i is around $\pm 1°$ and is eliminated by the accuracy of the experimental determination of the parameters k and ψ_0. The growth angle is constant ($\boldsymbol{\varepsilon} = 12° \pm 1°$ in the case of Si drop crystallization) at the planar isotropic growth front in the interval ($2\,\mathrm{mm} < r_i < 5\,\mathrm{mm}$). In the case of Ge drop crystallization the isotropic growth front was slightly convex (with a slope in $\alpha = 4°$ at the cone base, Fig. 6). These facts were taken into account in the calculations. Within the limits of experimental accuracy the calculated growth angle is $\varepsilon = 14° \pm 1°$ for Ge. The growth angle remains constant for the length ($2\,\mathrm{mm} < r < r_0 = 5\,\mathrm{mm}$). Under such conditions the parabola $z_1 = kr_1^2$ well approximates the cone profile near its base ($r \to r_0$), but the error increases near the cone apex ($r \to 0$). The calculated data cannot be taken into consideration for $0 < r < 2\,\mathrm{mm}$ (Fig. 14a and b).

In the case of InSb drop crystallization on a singular front $(\bar{1}\,\bar{1}\,\bar{1})$ the value of the growth angle is not constant and varies over the range $(24° \pm 1°) < \varepsilon < (30° \pm 1°)$. The explanation of such

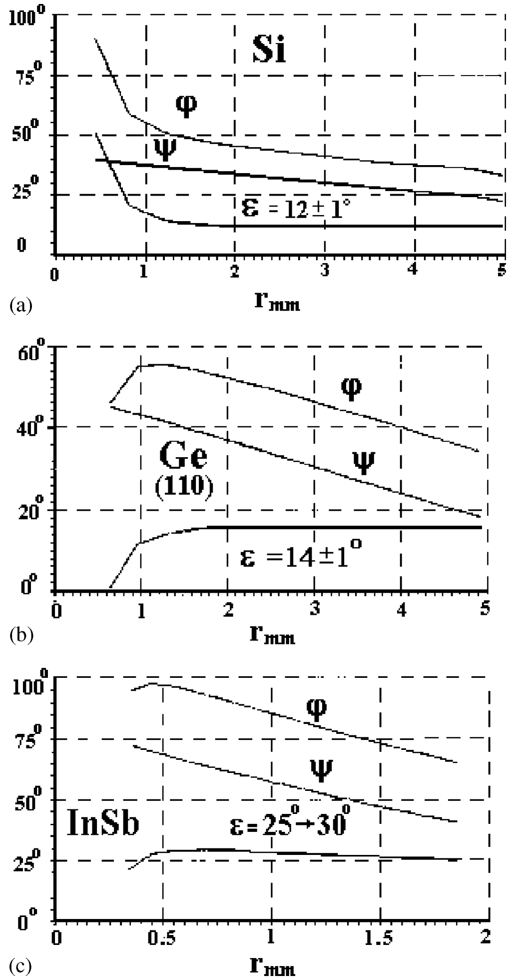


Fig. 14. The results of numerical calculations. (a) Cone on a isotropic Si growth front (see Fig. 15e), (b) cone Ge on a (1 1 0) on the growth front surface (see Fig. 15b) and (c) cone on a singular facet $(\bar{1}\,\bar{1}\,\bar{1})$ of InSb (Fig. 15d).

a feature of the InSb drop crystallization was given in Section 2 (case II, see Fig. 9).

The calculations, based on Eqs. (3.1)–(3.4), allows one to find the wetting angles of the crystalline surfaces by their own melt φ_{m0}. For germanium (Ge) $\varphi_{m0} = 30^\circ \pm 1^\circ$ and for silicon (Si) $\varphi_{m0} = 33^\circ \pm 1^\circ$ were obtained for case of the isotropic solidus-liquidus growth front surfaces. In the case of indium antimonide (InAs) $\varphi_{m0} = 62^\circ \pm 1^\circ$ was obtained for facet $(\bar{1}\,\bar{1}\,\bar{1})$ on the crystallization front.

4. Determination of the growth angle with assumption of its constancy

Now we pose the inverse problem. Namely, we try to calculate the cone shape and to compare it with its real shape. Initially we do not use any approximations, but suppose the constancy of the growth angle and the macroscopic front curvature. The solution of this problem is based on using two conservation laws: (i) the overall mass balance $dM = dm + d\mu = 0$, where dm is the incremental value of the crystal mass and $d\mu$ is the decrease of melt mass in the drop and (ii) the condition of growth angle constancy $\varepsilon = \varphi - \psi = \text{const}$. From the first requirement follows $\rho_S\, dV_S = -\rho dV$. Since $dV_S = \pi r^2(z)\, dz$, finally we have: $dV/dz = -\eta\pi r^2(z)$, where $\eta = \rho_S/\rho$. The second requirement gives the second differential equation $dr/dz = -ctg(\psi)$. For this set of simultaneous equations the initial conditions are the following: radius of the cone base is $r(0) = r_0$ and its initial volume $V(0) = \pi r_0^3 f(\varphi_0)(3 + f^2(\varphi_0)/6$, defined for the initial angle φ_0 (in this case identical with the wetting angle). The function $f(\varphi)$ was mentioned above (see Eq. (3.4)).

It is convenient to represent the obtained equations in dimensionless form using the following variables $\bar{r} = r/r_0$, $\bar{z} = z/r_0$ and $c = 3V(t)/\pi r^3$. Differentiating the last relation $dV/dz = r^2 c(dr/dz) + r^3(dc/dz)/3 = -\eta r^2$ and taking into account that $dr/dz = -ctg(\psi)$ we obtain the following pair of simultaneous equations:

$$\frac{dc}{d\bar{z}} = \frac{3}{\bar{r}}(cF(c) - \eta), \tag{4.1}$$

$$\frac{d\bar{r}}{d\bar{z}} = -F(c). \tag{4.2}$$

Here $F(c) = ctg\{2\,\mathrm{arctg}(\varphi) - \varepsilon_0\}$ and the initial conditions are: $\bar{r}(0) = 1$, $c(0) = 0.5f(\varphi_0)(3 + f^2(\varphi_0)$. The angle $\varphi = \varphi(c)$ can be obtained as the solution of the cubic equation (1.4) $f^3(\varphi) + 3f(\varphi) - 2c = 0$, for which the unique real solution is $f(\varphi) = tg(0.5\varphi) = [c + (1+c)^{1/2}]^{1/3} + [c - (1+c)^{1/2}]^{1/3}$ according to the Cardan formulae. The solution of the equations set (4.1), (4.2) one can write as the improper integrals [25]. In this article it is more convenient to use the results of

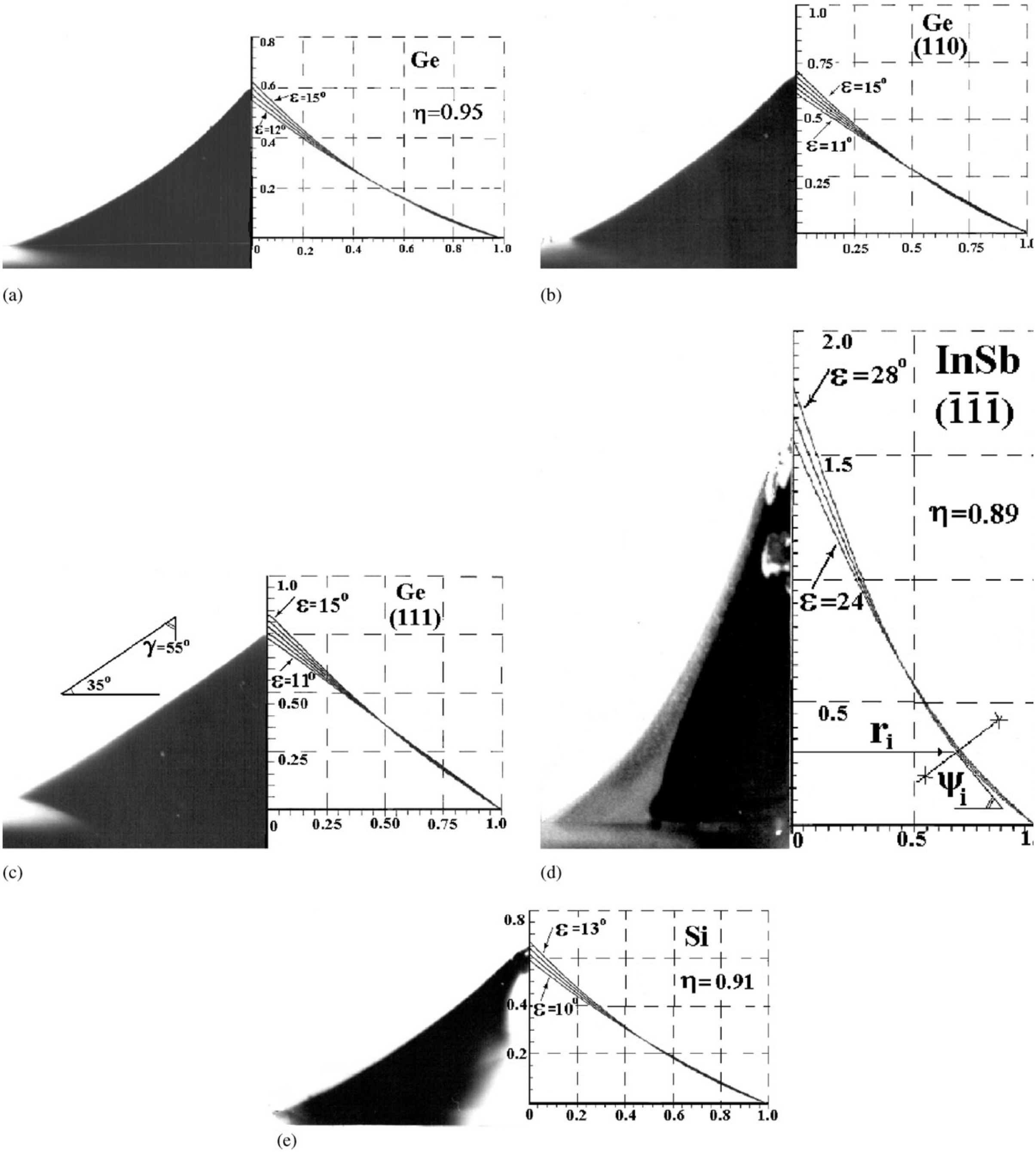


Fig. 15. The cones: (a) Ge on an isotropic growth front, (b) Ge on the growth front (1 1 0), (c) Ge on the growth front (1 1 1). In the case (c) the tilt of cone profile is constant. The slope angle at the cone apex is $2\gamma = 110°$; (d) InSb on the surface $(\bar{1}\,\bar{1}\,\bar{1})$ of the growth front, (e) **Si** on an isotropic growth front.

simple numerical calculations by the Runge–Kutta method.

The solutions of (4.1), (4.2) for a given material will depend on values of two parameters—the growth angle ε and the wetting angle φ_{m0}. Fig. 15 shows photos of cones (on the left) and calculated cone profile curves (on the right). The real and calculated cone shapes coincide well. This fact confirms the supposition of Section 4 about the growth angle constancy and the appropriateness of the approach of Section 3, based on parabolic approximations of the cone shape. However, the parabolic approximation is not always possible because the cone can have a more complicated shape. So the additional task of this section is the determination of the unknown parameters ε and φ_{m0} by use of features of cone geometry only.

One such cone feature is the unique dependence of the angle at the cone apex only from the relation of density ratio η and the growth angle (i.e. material properties of the crystal alone). The angle does not depend on the initial conditions of crystallization. This fact was mentioned first in Refs. [25,26]. The calculations of the slope angles $\gamma(r) = \varphi(r) - \varepsilon$ of the cone lateral surface, shown in Fig. 16, well illustrate such a cone feature. The calculated angle at the cone apex ($\gamma(\bar{r}\to 0) = \gamma_0$ is determined by choice of parameters η and ε only and not depend from the initial angle φ_0 [27].

Let us consider asymptotic solution of Eqs. (4.1) and (4.2), taking into account that always $\varepsilon \geqslant 0$, $\varphi \geqslant \varepsilon$ and $r \to 0$ when $z \to z_0$. After differentiating the expression for the drop volume $V = \pi r^3 f(\varphi)(3 + f^2(\varphi)/6$ and substituting in the

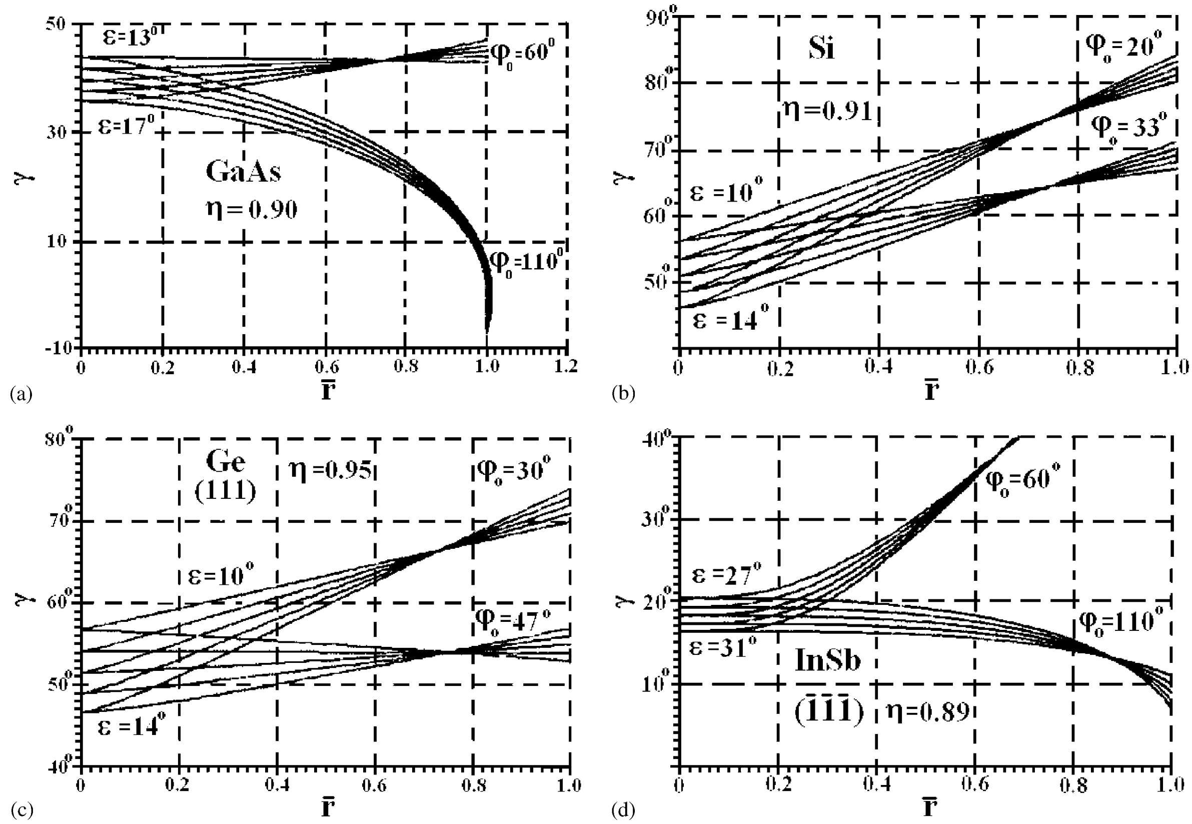
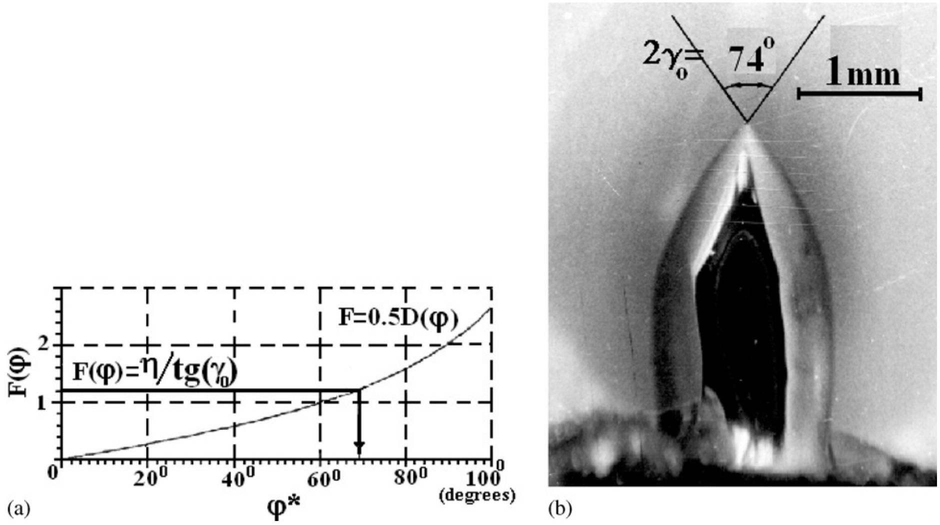


Fig. 16. Calculated values of the slope angles $\gamma(r) = \varphi(r) - \varepsilon$ of the lateral cone surface. The angle at the cone apex $\gamma(r \to 0) = \gamma_0$ does not depend on the initial angle φ_0 and is determined only by the parameter $\eta = \rho_S/\rho$ and the growth angle ε.



(a) (b)

Fig. 17. (a) Graphical solution of Eq. (4.4) (for a GaAs cone $\varphi^* = 69°$), (b) a GaAs cone. Note, this cone has no inflection point near the apex according to the theory (see text).

obtained expression $dr/dz = -ctg(\psi)$ we have

$$\frac{dV}{dz} = 0.5D(\varphi)r'_z - 0.5rf(\varphi) \times (1 + f^2(\varphi))\sin^{-1}(\varphi)\varphi'_z = -\eta, \quad (4.3)$$

where $D(\varphi) = f(\varphi)(3 + f^2(\varphi))$. Since for $r \to 0$ always $\varphi(r) > 0$, and so the second term in (4.3) tends to zero and can be neglected for the final stage of the melt drop crystallization on the cone apex.

Substituting the derivative $r'_z = -ctg(\psi)$ in the remaining part of expression (4.3), we obtain the following relation between the tangent slope angle to the meniscus (at the TPL) and the slope angle of the crystal lateral surface: $0.5D(\varphi)ctg(\psi) = -\eta$. Taking into account the relation $\psi = \varphi - \varepsilon$ and $\gamma = \pi/2 - \psi$, Eq. (4.3) can be rewritten as $0.5D(\varphi)ctg(\varphi - \varepsilon) = -\eta$ or

$$0.5D(\varphi)tg(\gamma) = \eta. \quad (4.4)$$

Eq. (4.4) allows one to determine the growth angle ε for a material with known η by using the measured angle γ_0 at the cone apex. The graphical solution of Eq. (4.4) (see Fig. 17a) allows one to determine the value φ^* which satisfies the relation $D(\varphi^*) = 2\eta/tg(\gamma_0)$. The value of the growth angle is determined from the expression $\varepsilon = \varphi^* - (90° - \gamma_0)$. Fig. 17b shows the cone after crystallization of a gallium arsenide melt drop. The measurements, based on a grown cone and its enlarged photo, give the angle at the cone apex $2\gamma_0 = 74°$ for the GaAs. The graphical solution of Eq. (4.4) for this angle gives the value $\varphi^* = 69°$ (Fig. 17a) and consequently the growth angle is $\varepsilon = 16°$ for GaAs. This value coincides with the experimental result in Ref. [26]. We note that the measurements of the cone angle at their apex do not present any difficulties; however, there exists the problem of the accuracy of such measurements. First of all, the measurement accuracy depends on the determination of the tangent to the lateral surface near the cone apex. The position of this tangent is subjective and its slope depends on the choice of the extent of the region selected.

The calculation results, shown in Fig. 16, allow one to estimate the optimum size of the selected region for reaching the required accuracy (for example $\pm 1°$). The curves $\gamma(r) = \varphi(r) - \varepsilon$ allow to determine, that such accuracy can be achieved by use of a region ($\sim 0.1r_0$) near the cone apex for materials of type GaAs and InSb, but for the Si and Ge it is necessary to choose considerably smaller sizes of the cone apex ($\sim 0.04r_0$) to reach such an accuracy. Table 2 show the results of measurement and calculation based on the experiment with crystallized drops on a growth front.

Table 2
The measurement and calculation results of the angles γ at the cone apex, the growth angles ε and wetting angles φ_0 based on experiments with melt drops crystallized on a crystal front

Material	$\gamma_{0(\mathrm{deg})}$	$\varphi^*_{(\mathrm{deg})}$	$\varepsilon_{\ (\mathrm{deg})}$	$\varphi_{0(\mathrm{deg})}$
Ge (isotropic)	44	60	14	30
Ge (1 1 0)	47	57	14	38
Ge (1 1 1)	55	49	12	47
InSb($\bar{1}\ \bar{1}\ \bar{1}$)	17	103	25→30	62
Si (isotropic)	47	55	12	33
GaAs	37	69	16	

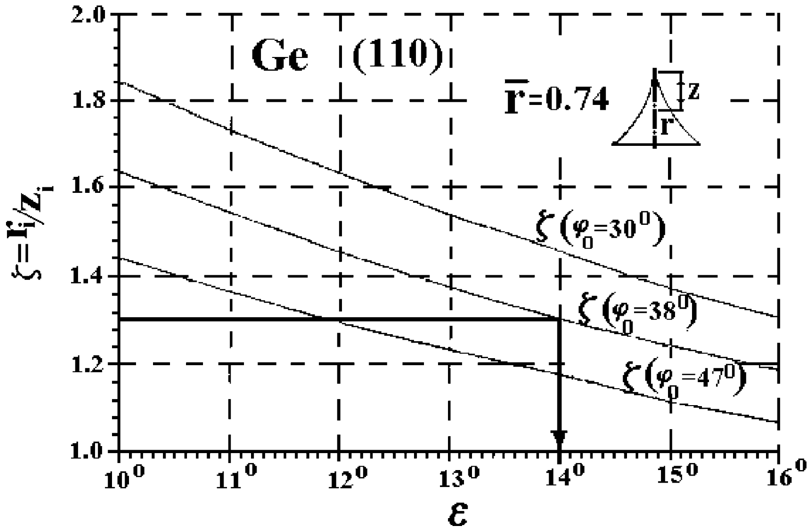


Fig. 18. The non-dimensional number $\xi = z_i/r_i$ allows one to calculate the wetting angle $\varphi_0 = 38°$ for materials with known growth angle ε (here, for a cone on the growth front (1 1 0) of Ge (see Fig. 15b)).

Note in the work [28] Eq. (4.4) is obtained on the assumption $d\varphi/dz = -d\gamma/dz = 0$. It means, that a cone apex always has a simple shape with a constant slope angle of its profile, like the Ge cone presented in Fig. 15c. Fig. 16 shows the limit of application of such an approach (see text above).

5. Determination of the wetting angle

Now we pose the task of determining the initial angle $\varphi(0) = \varphi_0$ to the melt drop at the TPL equal, in this case, to the equilibrium wetting angle φ_{m0} for materials with known value $\eta = \rho_S/\rho$ and the determined growth angle ε. It is possible to do by the use a few methods.

(A) For some arbitrary section of the cone $r_i/r_0(\leqslant 1)$ it is possible to compare the calculated non-dimensional number $\xi = r_i/z_i$ with the experimentally measured values.
(B) If the upper cone part does not allow one to carry out the measurement with the necessary accuracy, it is possible to use, in a similar manner, the values z_i from the cone base up to its ith section. Fig. 18 shows an example of the calculated parameter ξ, which allows one to determine the wetting angle $\varphi_0 = 38°$ for

germanium (Ge) with front orientation (1 1 0) (see also Fig. 15b and Table 2).

(C) It is possible experimentally to determine the slope of the lateral cone surface $\psi(\xi, \varepsilon)$ at some cone profile point r_i and then compare it with the calculated slope $\psi(\xi, \varepsilon, \varphi_0)$. To this end it is necessary to carry out the simple arbitrary geometrical cross-section of the cone profile curve (by use the enlarged cone photo) and then reconstruct the middle perpendicular to this cross-section (see Fig. 15d, on the right side). Thus, it is possible to determine the values of the angles ε and φ_{m0}.

In case of the Ge cone on a (1 1 1) surface, represented in Fig. 15c, the slope angle of the cone profile is constant $\gamma(r) = 55^\circ \pm 1^\circ$. It is not difficult to see from this and Fig. 16c, that only one of the calculated curves $\gamma(r, \varepsilon = 12^\circ, \varphi_0 = 47^\circ)$ can satisfy the requirement of profile slope constancy. The same growth angle $\varepsilon = 12^\circ \pm 1^\circ$ was determined for this Ge cone by the use of a simple method described in Section 3. The wetting angle for germanium front (1 1 1) is $\varphi_0 = 47^\circ \pm 1^\circ$ (Fig. 16c, Table 2).

Note that similarly, it is possible to determine the wetting and growth angles of the "usual" liquids (not melts). In experiments the liquid drop, situated on a solid substrate, could be slowly frozen beginning at its base [16] (Fig. 19). But it is necessary to note the difference in the experiments with the melt drops at a growth front and the drops of frozen fluids on a solid surface. The difference consists in the first moment of the liquid drop formation before the beginning of crystallization. The echelons of crystalline steps appears at the growth front and moves very quickly towards TPL (Fig. 1b) after separation of a crystal from the melt and shrinkage of a residuary thin film at the front into a melt drop.

This crystalline steps can be absorbed or accumulated by TPL. In the first case, the motion of TPL continues along the fresh crystalline surface. The front curvature of the drop base keeps invariable and the melt drop comes into the equilibrium position with the angle φ_0, equal to the wetting angle (Fig. 19b). In the second case, the local curvature of crystallization front in the vicinity of TPL can differ from its macroscopic curvature. In this case, the slope angle to the drop φ_0 at the TPL can be considered as the wetting

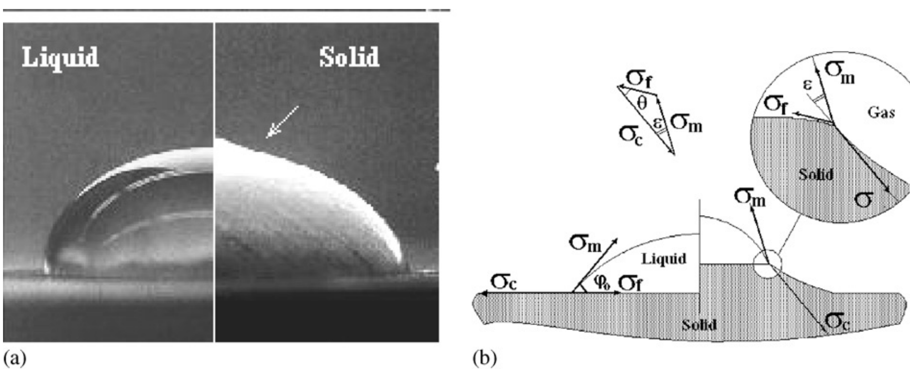


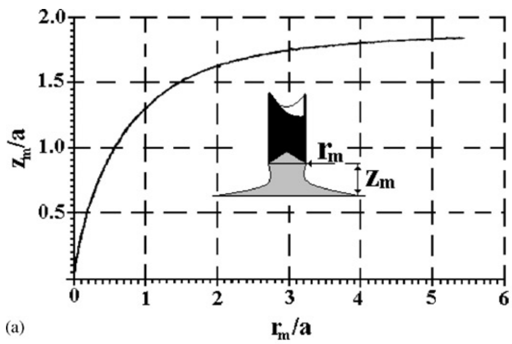
Fig. 19. (a) Photo (on the left) of a liquid water drop before the beginning of its crystallization by freezing (S.H. Devis, www.northwestern.edu). The shape of the frozen drop (on the right) very precisely coincides with the calculated shape (practically up to the cone top) for the growth angle $\varepsilon = 0^\circ$ [16]. It is possible to explain the presence of an inflection point near the cone apex (shown by the arrow) by transition from planar to the convex growth front at the end of the water drop crystallization. Compare: in Fig. 17b the GaAs cone without an inflection point near its apex according to the theory is shown. The liquid drop formation before freezing (a) differs from the initial formation of the melt drop (Fig. 6) at a growth front (see text). (b) Schematic of the equilibrium positions of three forces of interfacial tension σ_i, i=f, m, c at the TPL before and after beginning of the drop crystallization on the growth front. Vicinity of the TPL is shown enlarged. The triangle of the surface tension forces is closed $\sigma_c + \sigma_m + \sigma_f = 0$ and the growth angle ε is constant during drop crystallization.

angle φ_{m0}, strictly speaking, approximately. For liquid drops similar to the drops shown in Fig. 19a such problem does not appear.

6. Determination of the surface tension of the melt σ_m

Determination of the interfacial tension σ_m is especially difficult task in case of the high temperature melts. The known methods of experimental measurement of the tension or the value of capillary constant $a = (2\sigma_m/\rho g)^{1/2}$ (maximal pressure in bubble, methods involving a lying or a pendant drop, separation from a ring etc.), where g is the gravitational constant, are laborious and require special equipment [29]. However, a simple and accurate method for determination of the capillary constant a is based on the maximal separation meniscus height [14,15]. The measurement of these heights can be carried out by the use of a drive unit of the standard growth puller, moving a solid probe (with the known edge shape and size) used as the melt level sensor. This approach allow to determine the density of the melt (or other conductive fluids) ρ and calculate the unknown value of σ_m through the capillary constant a.

The method consists of the experimental determination of the non-dimensional ratio r_m/z_m of the typical solid probe size (for example its radius at the wetting perimeter r_m) to the maximal height of the meniscus, rising from the free melt surface z_m and following the solid probe. The maximal height of the meniscus is measured exactly at the moment of its breaking-off by an electrocontact method. Formally, the breaking-off occurs when the condition of positive definiteness of the second variation of the meniscus free energy functional $\delta^2 J \geqslant 0$ is violated. In detail, the problem of the meniscus stability was considered in [30]. Fortunately this variation of problem is simplified for a meniscus rising with a free liquid surface. In this case, the condition of positive definiteness of the functional $\delta^2 J \geqslant 0$ can be investigated simultaneously with the additional condition of the pressure constancy in the meniscus instead of the condition of its volume constancy. The calculation of maximal meniscus height is reduced to the solution of homogeneous Jacobi equation for selected extremal $r(z)$ (the meniscus profile). The envelope line (C-discriminant line) for the bundle of the extremal restricts the upper bound of the meniscus heights in this case. The careful numerical calculations of this task were carried out in [31,32]. This unique and one-valued curve $z_m(r_m)$ (Fig. 20a) is wonderful, since it restricts maximal heights *of all kinds of fluids*, including high-temperature melts in a dimensionless coordinates. In Fig. 20b a dimensionless ratio $\Omega(r/a) = r_m/z_m$, which used in determination of capillar constant, is shown. This function is well approximated by the equation $\Omega(r/a) = 0.4859(r/a) + 0.2474$ for $(r/a > 0.2)$. Here is used $\grave{a} = (\sigma_m/\rho g)^{1/2}$ similar to the data from [31]. A ratio $R/H = \Omega = r_m/z_m$ is especially simply determined for a solid probe with



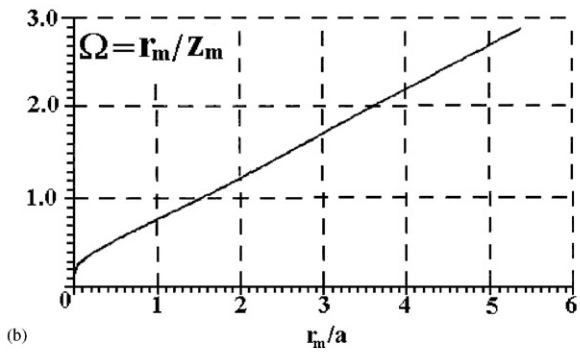


Fig. 20. (a) C-discriminant line in dimensionless coordinates (plotted according to the data from [31]) restricts maximal heights for all meniscus (any fluids), lifted after solid cylinder from a free liquid surface. (b) Graph of function $\Omega(r/a) = r_m/z_m$ used for the capillary constant determination. The last function can be described by equation $\Omega(r/a) = 0.4859(r/a) + 0.2474$ for $(r/a > 0.2)$.

known radius R of its cylindrical sharp edge (here H- is an experimental separation meniscus height). Using Fig. 20b (or the equation presented above) and experimentally determined value Ω one can definde the value r_m (or z_m) and then calculate the capillary constant $a = R/r_m$ (or $a = H/z_m$). Unknown surface tension $\sigma_{m=}\rho g a^2$ can be calculated for liquids with known density ρ.

Parameter ρ may be determined in the same experiment. This approach allows to investigate the dependence of σ_m on temperature and composition in the same experiment [14,15]. For improvement of measurement accuracy it is necessary to take into account the presence of the remaining drop on the probe end after meniscus breaking off, the probe edges sharpness and its orientation accuracy with respect to the melt surface.

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References

[1] V.V. Voronkov, Solid State Phys. 5 (2) (1963) 571.
[2] V.V. Voronkov, Sov. Phys. Crystallogr. 19 (1974) 922.
[3] V.V. Voronkov, Sov. Phys. Crystallogr. 23 (1978) 249.
[4] V.V. Voronkov, Bull. Russ. Acad. Sci., Ser. Phys. 44 (1980) 226.
[5] V.V. Voronkov, J. Crystal Growth 52 (1981) 311.
[6] V.V. Voronkov, Bull. Russ. Acad. Sci., Ser. Phys. 47 (1983) 210.
[7] V.V. Voronkov, Bull. Russ. Acad. Sci., Ser. Phys. 49 (1985) 2467.
[8] W. Bardsley, F.C. Frank, G.W. Green, D.T.J. Hurle, J. Crystal Growth 23 (1974) 341.
[9] G.A. Satunkin, J. Crystal Growth 154 (1995) 172.
[10] G.A. Satunkin, A.G. Leonov, J. Crystal Growth 102 (1990) 592.
[11] S.N. Rossolenko, J. Crystal Growth 231 (2001) 306.
[12] G.F. Bolling, W.A. Tiller, J. Appl. Phys. 31 (1960) 1345.
[13] J.K. Kristensen, R.M.J. Cotterill, Philos. Mag. 36 (1977) 437.
[14] G.A. Satunkin, B.S. Rd'kin, V.N. Kurlov, S.N. Rossolenko, V.A. Tatarchenko, Yu.A. Tuflin, Cryst. Res. Tecnol. 21 (8) (1986) 995.
[15] V.N. Kurlov, B.S. Rd'kin, J. Crystal Growth 104 (1990) 80.
[16] D.M. Anderson, M. Grae Worster, S.H. Davis, J. Crystal Growth 163 (1996) 329.
[17] E.I. Givargizov, Growth Whiskers and Lamellar Crystals from the Vapor, Nauka, Moscow, 1977, p. 303.
[18] G.A. Satunkin, V.A. Tatarchenko, Sov. Phys. Crystallogr. 30 (1985) 772.
[19] A.A. Chernov, Sov. Phys. Usp. 73 (1963) 277.
[20] C. Herring, in: The Physics of Powder Metallurgy, McGraw-Hill, NY, 1951, p. 143.
[21] C. Herring, Phys. Rev. 82 (1) (1951) 87.
[22] D.W. Hoffman, J.W. Chan, Surf. Sci. 31 (1972) 368.
[23] V.V. Voronkov, Sov. Phys. Crystallogr. 12 (1967) 831.
[24] V.V. Voronkov, Sov. Phys. Crystallogr. 19 (1974) 228.
[25] A. Sanz, J. Crystal Growth 74 (1986) 642.
[26] V.A. Antonov, V.V. Selin, Electr. Techn. Mater. 3 (152) (1981) 20 (in Russian).
[27] G.A. Satunkin, in: V.P. Ginkin (Ed.), Proceedings of the IV International Conference ICSC-2001, Vol. 2, Obninsk, Russia, September 24–28, 2001, p. 377.
[28] V.A. Antonov, V.V. Selin, J. Crystal Growth 241 (2002) 507.
[29] A.W. Adamson, Physical Chemistry of Surfaces, III Edition, Wiley, NY, 1976, p. 568.
[30] A.V. Zdanov, G.A. Satunkin, R.P. Ponomareva, J. Colloid Interface Sci. 104 (2) (1985) 334.
[31] W. Uelhoff, K. Mika, Julich-Report No. 1195-FF, 1975.
[32] K. Mika, W. Uelhoff, J. Crystal Growth 30 (1975) 9.